



Department of Computer Science & Engineering
Chitkara University Institute of Engineering & Technology
(Accredited by NAAC with Grade 'A+')

Ref No: CUIET/CSE/ACAD/2024/472

Date: 21st October 2024

NOTICE

ATTENTION: - B.E CSE Batch 2022

Test Type: ST **Test No:** 1 **Total Marks:** 40 **Duration:** 90 minutes

Name of Course Coordinator: Dr. Astha Gupta

Course Code: 22CS025 **Course Title:** Algorithm Design & Implementation

Day: Saturday **Date:** 26-10-2024 **Time:** 09:00 AM – 4:00 PM

Mode: Online on Campus **Semester:** B.E. (CSE) & 5th Sem **Platform:** Testpad

Syllabus for Test: (As Per CHO-Lecture No. 1-40) Concept of Algorithm complexity- Big O notation, Algorithm complexity -Average, best and worst case. Practice Problem: Prime Factorization, GCD of two numbers, Distribute in circle. Introduction to Hash Table; Collision, Collision Avoidance Strategies- Linear Probing, quadratic probing, Separate Chaining. Practice Problem: Noise In the Library, try balancing the scale, Find out the winner. Heap and Priority Queues: Heap Sort, Implement Priority Queue using Linked List. Practice Problem: Find max/min in the continuous stream of data, sort an array using heap sort, Check if a given tree is max-heap or not. Introduction to Binary Tree, Creating Binary Tree, Tree Traversal. Binary Search Trees: Insertion and Deletion, Binary Search Trees Traversals. Practice Problem: Find the lowest common ancestor of a given two nodes in a binary search tree, Find the kth smallest element in the binary search tree. Divide and Conquer Strategies: Binary search Algorithm, Merge Sort, Quick Sort and Analysis. Practice Problem: Count Inversions, Find frequency of each element in a limited range array in less than $O(n)$ time. Greedy Algorithms: Fractional Knapsack problem. Practice Problem: Interval scheduling, Job Scheduling with deadlines. Backtracking: Introduction, N-Queen Problem. Practice Problem: Robot Movement, Solve Sudoku, Rat in a Maze, Print all strings of n-bit.

Format of Assessment:

1 Mark MCQ - 10: 10 Marks

2 Marks MCQ – 5: 10 Marks

5 Marks Coding – 2: 10 Marks

10 Marks Coding – 1: 10 Marks

Signature of Course Coordinator: